VENUS

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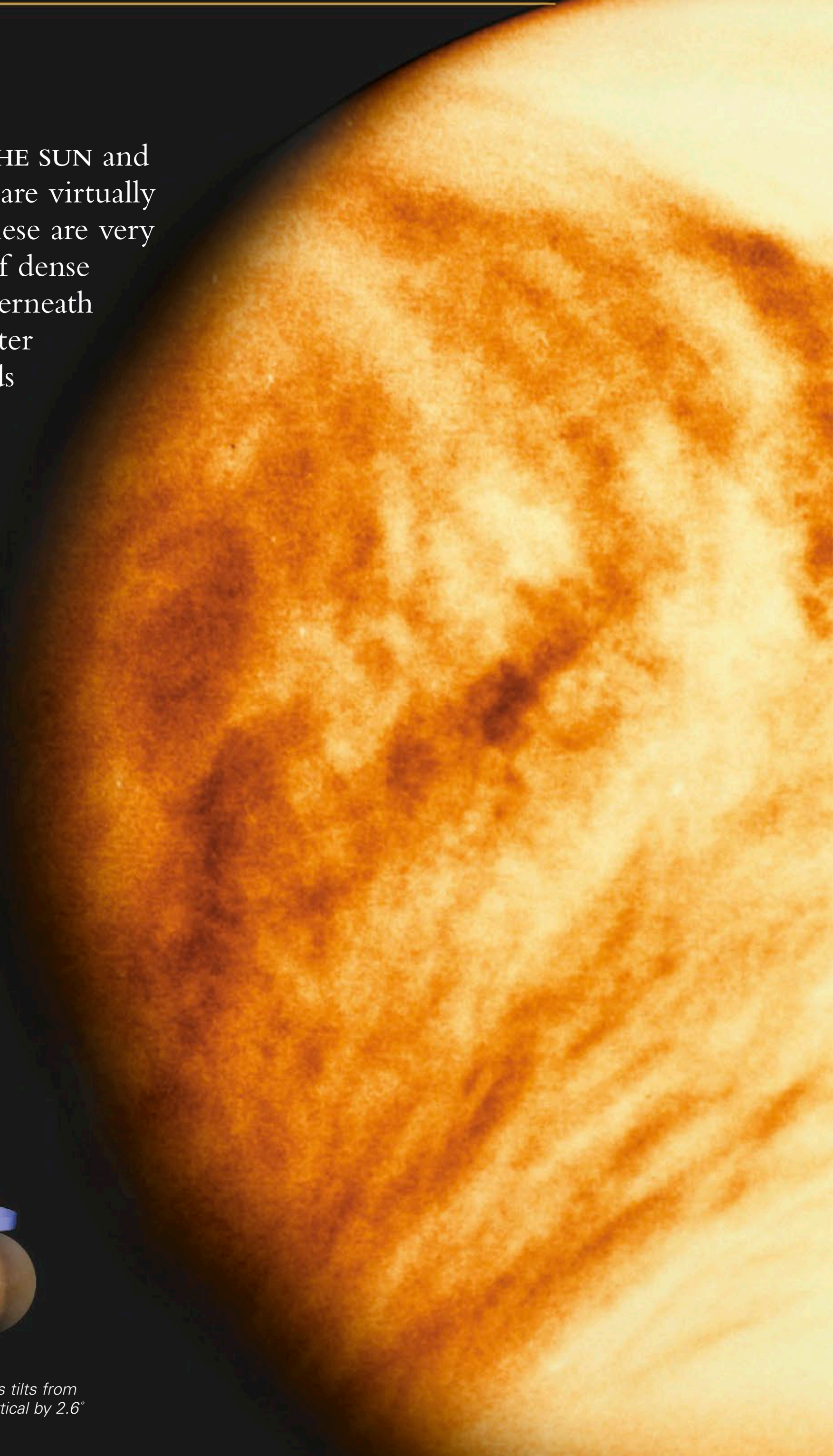
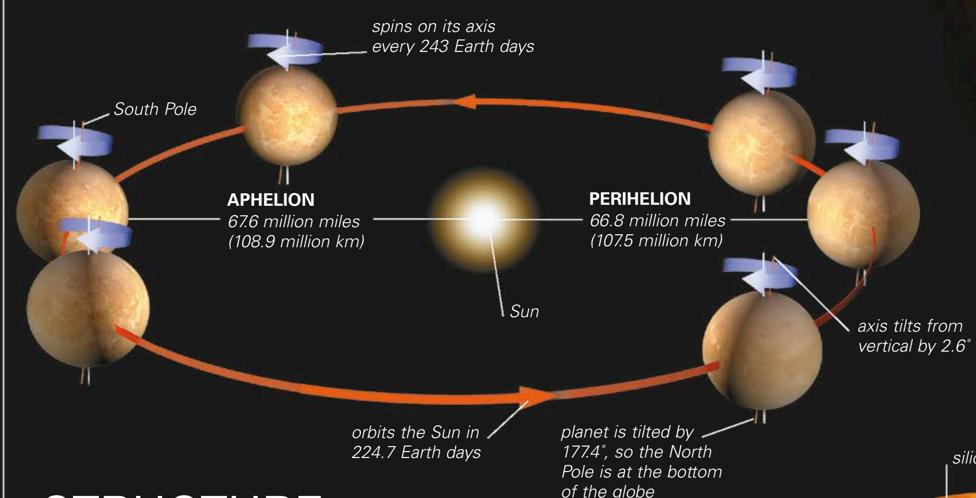


VENUS IS THE SECOND PLANET FROM THE SUN and Earth's inner neighbor. The two planets are virtually identical in size and composition, but these are very different worlds. An unbroken blanket of dense clouds permanently envelops Venus. Underneath lies a gloomy, lifeless, dry world with a scorching surface, hotter than that of any other planet. Radar has penetrated the clouds and revealed a landscape dominated by volcanism.

ORBIT

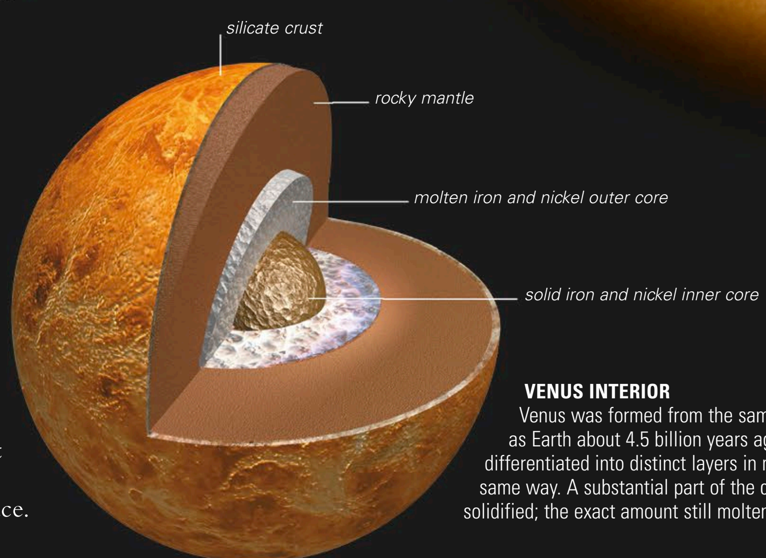
Venus's orbital path is the least elliptical of all the planets. It is almost a perfect circle, so there is little difference between the planet's aphelion and perihelion distances. Venus takes 224.7 Earth days to complete one orbit. As it orbits the Sun, Venus spins extremely slowly on its axis—slower than any other planet. It takes 243 Earth days for just one spin, which means that a Venusian day is longer than a Venusian year. However, the time between one sunrise and the next on Venus is 117 Earth days. This is because the planet is traveling along its orbit as it spins, so any one spot on the surface faces the Sun every 117 Earth days. Venus's slow spin is also in the opposite direction from most other planets. Venus does not have seasons as it moves through its orbit. This is because of its almost circular path and the planet's small axial tilt. Venus's orbit lies inside that of the Earth, and about every 19 months, Venus moves ahead of Earth on its inside track and passes between our planet and the Sun. At this close encounter, Venus is within 100 times the distance to the Moon.

SPIN AND ORBIT
Venus is tipped over by 177.4°. This means its spin axis is tilted by just 2.6° from the vertical. As a result, neither of the planet's hemispheres nor poles points markedly toward the Sun during the course of an orbit.



STRUCTURE

Venus is one of the four terrestrial planets and the most similar of the group to Earth. It is a dense, rocky world just smaller than Earth and with less mass. Venus's Earth-like size and density leads scientists to believe that its internal structure, its core dimensions, and the thickness of its mantle are also similar to Earth's. So Venus's metal core is thought to have a solid inner part and a molten outer part, like Earth's core. In contrast to Earth, Venus has no detectable magnetic field. The planet spins extremely slowly compared to Earth, far too slowly to produce the circulation of the molten core that is needed to generate a magnetic field. Venus's internal heat—generated early in the planet's history and from radioactive decay in the mantle—is lost through the crust by conduction and volcanism. Heat melts the subsurface mantle material, and magma is released onto the surface.



VENUS INTERIOR
Venus was formed from the same material as Earth about 4.5 billion years ago and has differentiated into distinct layers in much the same way. A substantial part of the core has solidified; the exact amount still molten is unknown.